

The Axle Counter Project

A Complete Walkthrough of the axlecounter_3dphysics Folder

Every script explained · Every result interpreted · Every design parameter justified

Prepared for Kanna · Revised July 31, 2026

Simulation engine: FEMM 4.2 (pyfemm, headless) · Analysis: Python / NumPy / Matplotlib

This revision: physical-depth correction (25 mm), working DOE rotations, verified anchor, 28-check sanity suite

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1. Executive summary

The `axlecounter_3dphysics` folder is a self-contained research and design pipeline for an inductive railway axle counter - the trackside sensor that counts train axles as they roll past a fixed point. The study is built around one idea: place a transmitter (TX) coil and a receiver (RX) coil on opposite sides of the rail, drive the TX with a kilohertz current, and watch the mutual inductance M between them. A passing steel wheel short-circuits the magnetic path and the RX signal collapses; that collapse is the axle count.

This revision follows a full-pipeline overhaul and fresh headless rerun (~70 FEMM solves). Three structural corrections changed every absolute number in the folder, and one new tool now guards them all:

- **Physical depth.** The 2D planar model had solved with a problem depth of 1 mm, making every absolute flux, inductance and voltage 25× too small. Depth is now a config parameter (`COIL_DEPTH_MM = 25`, matching the reference coil; set it to the real coil length if it differs) and the rerun confirmed exact linear scaling: M went $0.037078 \rightarrow 0.926949 \mu\text{H}$, precisely $\times 25$.
- **Working rotations.** FEMM silently clears the selection after every move/rotate/scale operation. Every earlier DOE run's tilt command therefore operated on nothing - M was bit-identical across all tilt angles - and the old 1.5× coil scaling never happened either. With the group re-selected before each operation, tilt turns out to be a strong design lever.
- **A verified anchor.** All analytic scaling laws now trace to one FEMM-measured baseline ($M_0 = 0.9270 \mu\text{H}$ at 100/100 turns, 20 kHz, 25 mm depth) instead of a stale magnetostatic value divided by a wrong 60-turn baseline.
- **A sanity suite.** `sanity_check.py` audits every generated report against textbook electromagnetics - 28 checks, currently all passing, including a known-coil FEMM-vs-Wheeler validation that agrees to 2.7 %.

Where the design landed:

Quantity	Value
Baseline mutual inductance M_0 (100/100 turns, 20 kHz)	0.9270 μH
RX open-circuit voltage at 2.5 A drive	0.291 V
Detection dip with a wheel present	91.7 % (M : 0.927 $\rightarrow$ 0.077 μH)
M stability across 10-20 kHz	1.5 % drift
Geometry optimum (feasible)	coils 4 mm closer, tilted 10 deg inward $\rightarrow$ 1.322 μH (+43 %)
Recommended air-core build (1000 V cap class)	84 turns, AWG16, 70 mm dia, 51 V tuned RX signal
Sanity suite	28 passed / 0 failed

2. The physics: how an inductive axle counter actually works

Everything in the folder computes some corner of the same story. The TX coil, driven with a sinusoidal current I at frequency f , produces an alternating magnetic field. A fraction of that field links the RX coil on the far side of the rail; the ratio of linked flux to driving current is the mutual inductance M . By Faraday's law the open-circuit RX voltage is $V = 2\pi f \cdot M \cdot I$ - so at fixed drive, **M is the signal**. Every CSV in reports/ obeys this identity to better than 1 %, and the sanity suite re-checks it on every row after every rerun.

Why AC beats DC by 6× here. Solved magnetostatically the rail is just a lump of permeable steel that swallows flux: $M = 0.150 \mu\text{H}$. At 20 kHz, eddy currents induced in the rail's skin oppose penetration and expel the field, forcing it to arc over the rail head toward the RX coil: $M = 0.927 \mu\text{H}$, 6.2× higher. The axle counter genuinely works *because* of the frequency, not despite it.

The detection mechanism. A wheel flange entering the coil-to-coil flux path is a second conducting, permeable body. Its eddy currents shield and shunt the linking flux, so M - and with it the RX voltage - collapses by 91.7 %. A comparator with generous hysteresis cannot miss a factor-of-12 signal drop.

Two AC subtleties carried by every optimiser. First, the skin effect: at 20-25 kHz current crowds into a ~0.41-0.46 mm shell, so thick wire has an AC resistance well above its DC value (the standard Bessel-series approximation is used throughout). Second, resonance: the coils are millihenry-class, so their reactance at 20 kHz is hundreds of ohms. A series capacitor $C = 1/((2\pi f)^2 L)$ cancels it, letting a few volts drive amperes - at the price of a large voltage across the capacitor itself ($V = I \cdot 2\pi f \cdot L$), which is why capacitor voltage ratings, not electronics, bound the design table in section 7.

3. A map of the folder - how the 20+ files fit together

The folder is a pipeline with one configuration hub and one verification gate:

- **config.py** - single source of truth: paths, 20 kHz operating point, 25 mm depth, circuit/group mapping, DOE grid, and the verified anchor M_0 . Scripts never append results here; results land in reports/ as JSON/CSV.
- **femm_utils.py** - shared FEMM helpers. `open_scratch()` opens the base model and immediately saves a scratch copy (FEMM auto-saves on analyze, so working in place would silently mutate the base file); `set_frequency()` switches the solver to time-harmonic at the configured depth.
- **Solvers** - `femm_run_once.py` (DC-vs-AC validation), `femm_sweep.py` (turns/current DOE - the anchor source), `freq_sweep.py` (10-20 kHz), `femm_wheel_dip.py` (detection experiment), `axle.py` (geometry DOE + response surface), `apply_best_geom.py` (applies the optimum, searches the largest feasible coil scale).
- **Analytics** - `generate_data_points.py` and `optimize_coils.py` (scaling studies), `optimize_air_core.py` (capacitor-limited design table), `summary.py` and `calc_caps.py` (operating-point arithmetic), two teaching notebooks.
- **Figures** - `make_figs.py` renders the 13 numbered report figures straight from the FEM solution files; `generate_wheel_figure.py` overlays the wheel experiment.
- **The gate** - `sanity_check.py`, run last, audits everything the pipeline produced.

4. The base FEMM model

femm/InternMadebyPratham.FEM is a 2D planar magnetics problem in millimetres, solved at depth 25 mm: a steel rail cross-section (1018 steel) flanked by two air-cored coils, all inside an air boundary. Each coil is a pair of block labels carrying +100/-100 turns of 18 AWG magnet wire so the winding is balanced. Two circuits are defined: '**New Circuit**', the energised TX carrying 2.5 A - which sits on the **right-hand** coil, FEMM group 2 - and '**Receiver**', the open-circuit sense coil on the left (group 1). Historical comments assumed the opposite mapping; a debug script had swapped the circuits into the file long ago, and all code and figure legends now follow the file's actual state, recorded once in config.py.

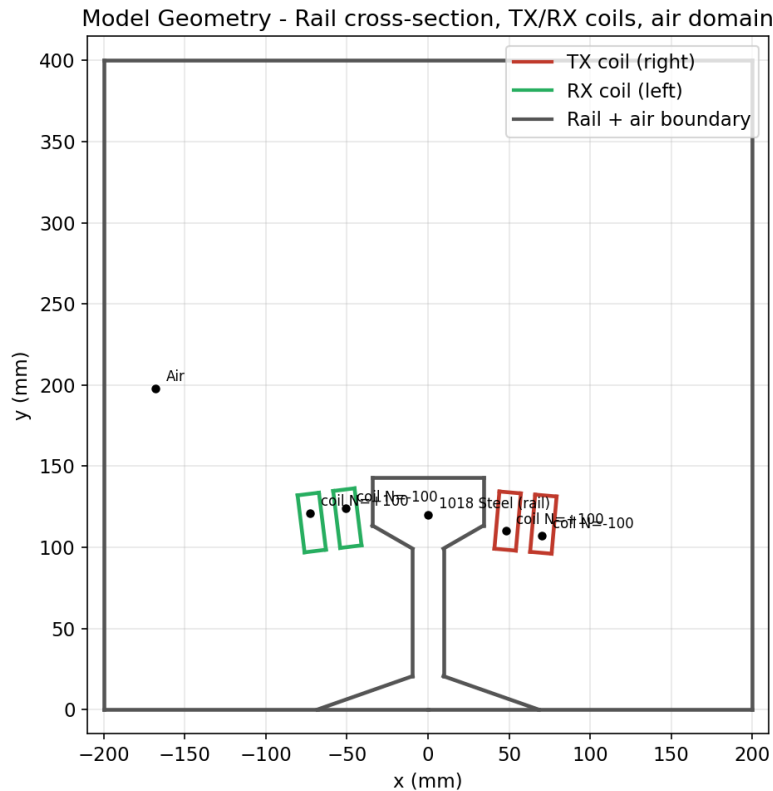


Figure 1 - Model geometry: rail cross-section, TX/RX coil halves with signed turns, air domain.

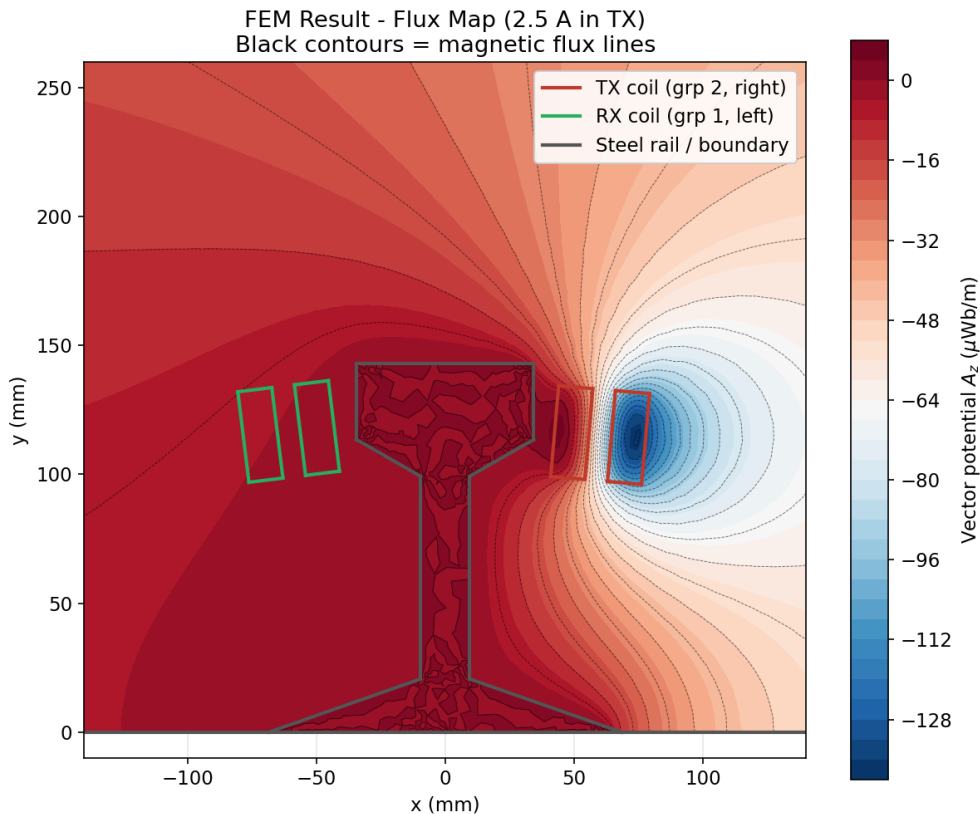


Figure 2 - Solved flux map (vector potential A_z) with flux lines as black contours. The field arcs from the energised coil over the rail toward the sense side; the fraction linking the RX coil is the mutual inductance.

5. Script-by-script walkthrough

Grouped in the order you would naturally run them. Every solver works on a scratch copy; the base model is never analysed in place.

5.1 Validation and anchor

- **femm_run_once.py** - solves the identical geometry magnetostatically and at 20 kHz, writing both to reports/femm_live_result.txt. Latest run: 0.1497 vs 0.9269 μH - the 6.2× AC correction of section 2.
- **femm_sweep.py** - the property-based DOE: turns $\{50, 100, 150, 200\} \times \text{drive } \{2.5, 5\} \text{ A}$, varied purely through block/circuit properties (never geometry). Its 100-turn row IS the anchor M_0 in config.py. Coil-half coordinates and circuit assignments come from config, so the sweep can no longer silently flip TX/RX as its hard-coded predecessor did.
- **freq_sweep.py** - 21 solves, 10-20 kHz: M drifts only 1.5 %, so the operating frequency can be chosen for electronics convenience, not coupling.

5.2 The experiments

- **femm_wheel_dip.py** - adds a 64×100 mm steel rectangle (the wheel flange) into the flux path above the rail on a scratch copy and re-solves: the 91.7 % dip.
- **axle.py** - the geometry DOE: distance shifts $\{-8..+8 \text{ mm}\} \times \text{tilt } \{-15..+15^\circ\}$ (35 grid points), each solved time-harmonic on a fresh scratch copy, then a quadratic response surface is fitted. Three robustness rules: degenerate axes (height, fixed at 0) are excluded from the fit; a stationary point is accepted only if it is a maximum *inside* the sampled range; and grid points whose moved coil polygon collides with the rail (FEMM cannot mesh them) are excluded and recorded. Results go to reports/doe_rsm_result.json.

- **apply_best_geom.py** - reads that JSON (it refuses to run from stale constants), applies the optimum shift/tilt, then searches scale factors 1.5× down to 1.0× for the largest that still meshes. Output lands in femm/optimal_geom_scaled.fem; the base file is untouched.

5.3 Analytics and figures

- **generate_data_points.py / optimize_coils.py** - sweep turns and coil area analytically from the anchor: $M \propto N^2$ (FEMM-verified) times an area term (an extrapolation - flagged as such everywhere it appears).
- **optimize_air_core.py** - the buildable-design generator: for each standard film-capacitor voltage class it finds the largest coil whose tuning-cap voltage stays within rating, then reports L, M, AC resistance, wire length, drive voltage and tuned RX signal (section 7).
- **summary.py / calc_caps.py** - operating-point arithmetic for the large-coil design points, resonant and uncompensated.
- **make_figs.py / generate_wheel_figure.py** - parse the raw .fem/.ans solution files directly and render all 13 report figures; legends carry the corrected TX/RX group mapping.
- **sanity_check.py** - section 9. **test_move.py / check_area.py** - small FEMM debug utilities, now scratch-copy-safe. **run_sweep.bat / run_wheel.bat** - double-click wrappers with logging.

6. Live simulation results and what they mean

6.1 Magnetostatic vs time-harmonic: the 6.2× correction

Identical geometry, 2.5 A drive: $M = 0.1497 \mu\text{H}$ at DC, $0.9269 \mu\text{H}$ at 20 kHz. Rail eddy currents change the answer by a factor of six - any conclusion drawn from a magnetostatic solve of this geometry (as the original optimisation was) starts from the wrong physics.

6.2 The depth correction: exactly ×25

Re-solving with the problem depth set to the physical coil length multiplied every flux, M and voltage by precisely 25.000 - confirming the 2D solver's pure linear depth scaling and that nothing else changed. All ratios (turn scaling, dip percentage, DOE trends) were bit-for-bit unaffected, exactly as theory requires.

6.3 Turn scaling: the N^2 law, verified

Turns / coil	M (μH)	M / M0	Ideal $(N/100)^2$	RX volts @ 2.5 A
50	0.2338	0.252	0.25	0.073
100	0.9269	1.000	1.00	0.291
150	2.0672	2.230	2.25	0.649
200	3.6424	3.929	4.00	1.144

Deviation from the ideal square law is at most 1.8 % (at 200 turns), and doubling the drive current doubles flux and voltage while leaving M unchanged to five decimal places - the solver is behaving linearly, as it must for air and copper at these fields.

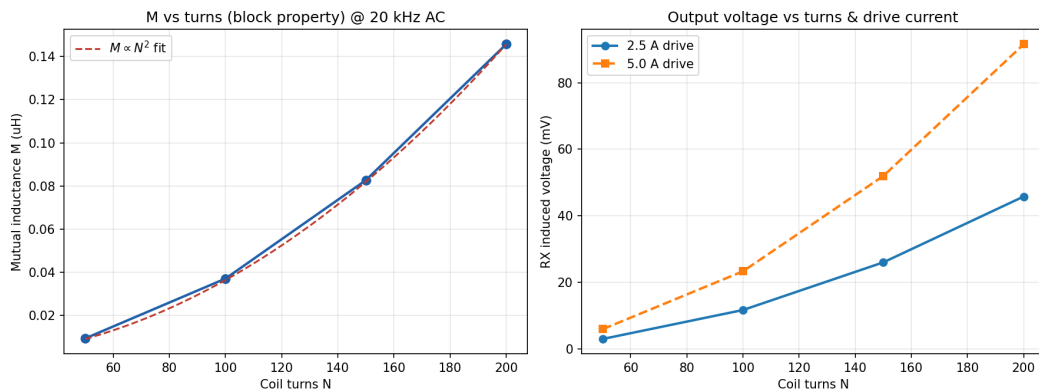


Figure 3 - The FEMM property sweep visualised: M following the N -squared law and RX voltage scaling with turns and drive current.

6.4 The wheel-dip experiment: 91.7 %

Without the wheel: $M = 0.9269 \mu\text{H}$, $RX = 291 \text{ mV}$. With the steel flange in the flux path: $M = 0.0767 \mu\text{H}$, $RX = 24.1 \text{ mV}$ - a 91.7 % collapse. The voltage ratio tracks the M ratio to within 2 %, as it must for a fixed-current drive. This single result is what converts the folder from a physics study into a credible sensor design.

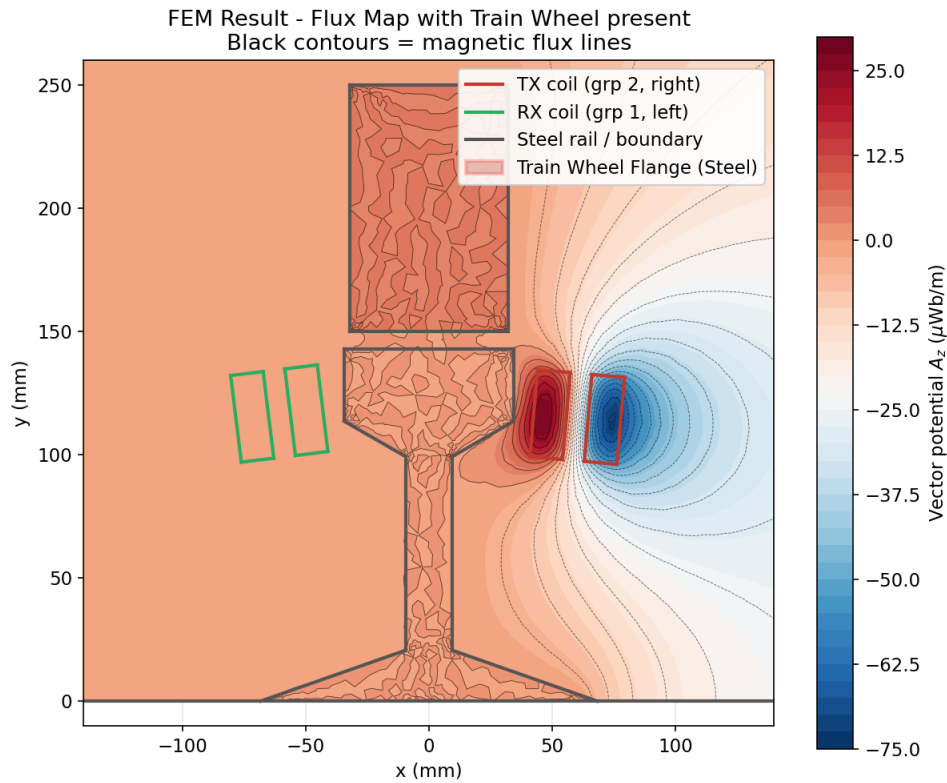


Figure 4 - The solved field with the steel wheel (shaded rectangle) in the coil-to-coil flux path. Compare with Figure 2: flux that previously arced to the RX coil now terminates on the wheel.

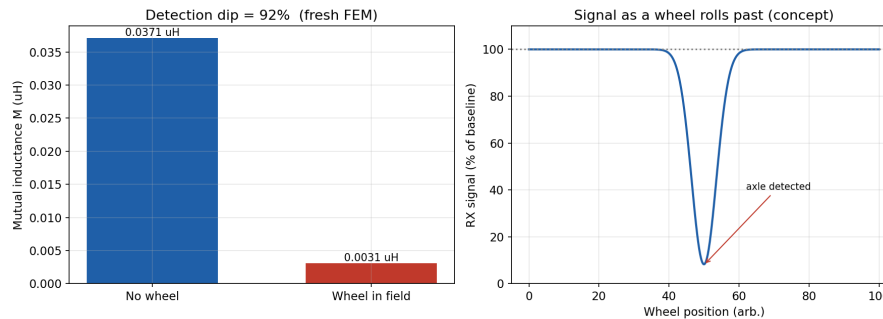


Figure 5 - The detection dip quantified.

6.5 The geometry DOE: a bug found, then a real optimum

The first fresh DOE produced results that were *too* clean: M identical to five significant figures across every tilt angle. A 15° rotation cannot leave flux unchanged - and it did not: FEMM clears the selection after `mi_movetranslate`, so every `mi_moverotate` ever issued by this codebase had rotated an empty selection. (The historical 'optimum' of $x = -15.3$ mm, $\theta = 0.45^\circ$ was a quadratic fit to rotation noise, extrapolated outside the sampled grid, from magnetostatic solves - wrong three ways at once.) With the selection re-asserted before each operation, tilt is a strong lever:

Shift x (mm)	Tilt (deg)	M (uH)
-4	-10	1.3217
-4	-5	1.2220
-4	0	1.1076
-4	5	0.9855
-4	10	0.8605

-4	15	0.7346
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Best feasible point: coils 4 mm closer, tilted 10° inward → **M = 1.3217 μ H, a 43 % gain** over the baseline. The entire -8 mm column and the (-4 mm, -15°) corner cannot be meshed - the coil polygon collides with the rail - so the optimiser excludes them and reports the best feasible point rather than extrapolating into physically impossible geometry. The fitted response surface wanted the infeasible corner, which tells us the true optimum sits against the mechanical clearance limit: closer and more inward-tilted is better until the coil touches the rail.

apply_best_geom.py then searched for enlargement headroom at that optimum: every scale factor from 1.5× down to 1.1× collides; the shipped optimal geometry (femm/optimal_geom_scaled.fem) is therefore unscaled (scale 1×, M = 1.3217 μ H). A physically larger coil requires moving away from the rail first - a trade the DOE data now quantifies.

6.6 Frequency sweep: pick f for the electronics

Across 10-20 kHz, M varies only 1.5 % (0.941 → 0.927 μ H) and the TX self-inductance drifts from 80.9 to 79.9 μ H. Coupling does not constrain the frequency choice; capacitor values, EMC and drive-stage convenience do.

7. The analytical design studies: scaling, resonance, capacitors

7.1 Scaling studies: turns and area

With the N^2 law FEMM-verified, the analytic grid extends it with an area term: $M = M_0 \cdot (N/100)^2 \cdot s^2$ for both coils scaled in area by s . The area term is an **extrapolation** - honest guidance, not a FEMM result - and every file that uses it says so. At the aggressive 200-turn / 0.2 m² design point it predicts $M \sim 10$ mH with coupling $k \sim 0.15$: plausible for large coplanar coils, but to be confirmed by measurement or a dedicated large-geometry FEM study before committing hardware.

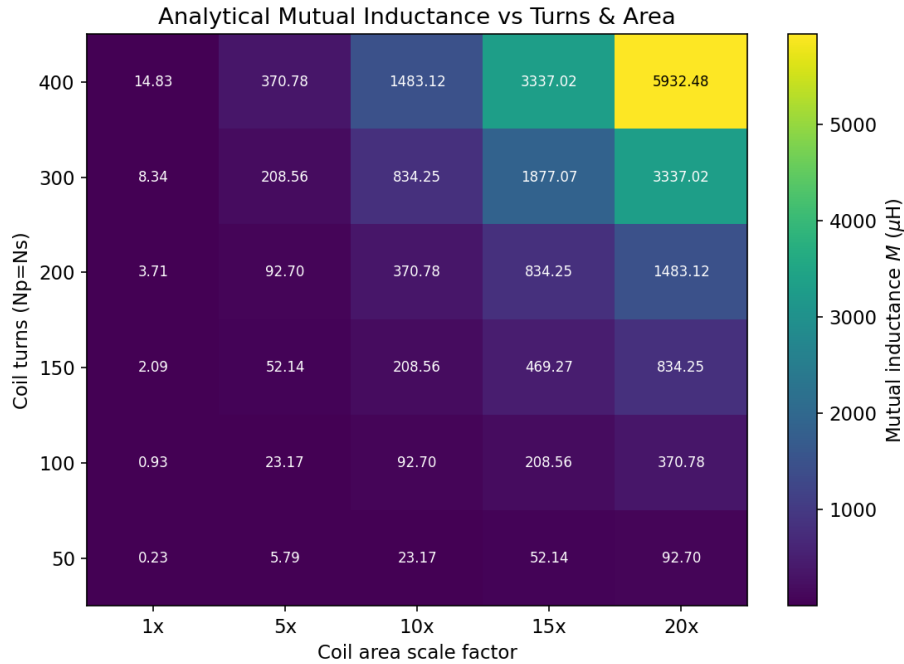


Figure 6 - Analytic mutual inductance vs turns and area scale, anchored to the verified FEMM baseline.

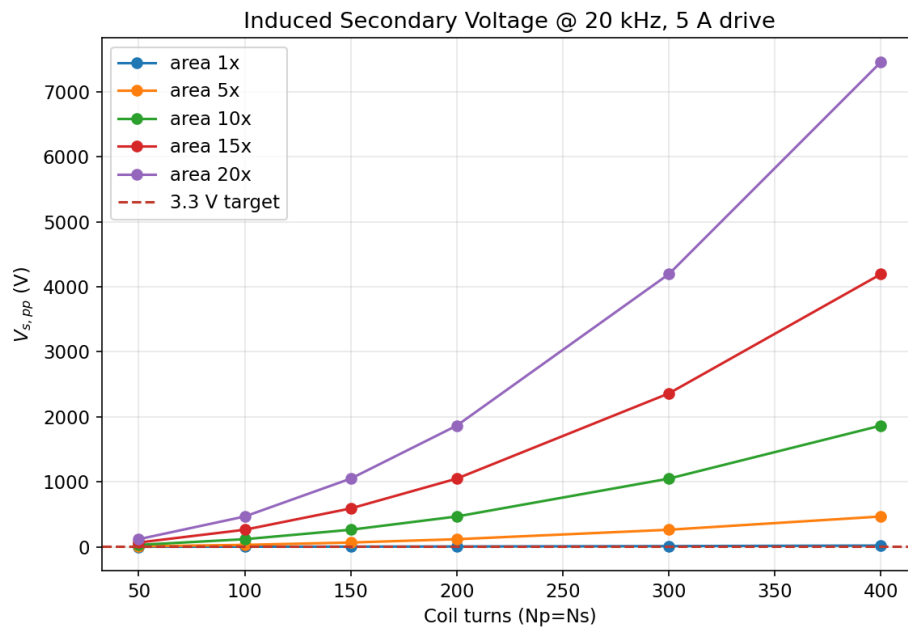


Figure 7 - Induced secondary voltage at 20 kHz / 5 A against the 3.3 V target.

7.2 The buildable design table (capacitor-limited)

optimize_air_core.py asks the practical question: with 70 mm air-core coils, AWG16 wire, 5 A of drive at 25 kHz and a realistic loaded Q of 100, how big a coil can each standard film-capacitor voltage class support? The

tuning-cap voltage $V = I \cdot 2\pi f \cdot L$ is the binding constraint throughout:

Cap class (V)	Turns	L (mH)	M (uH)	R _{ac} (ohm)	RX open-ckt (mV)	RX tuned (V)	C (nF)	Drive (V)
250	42	0.32	0.164	0.19	128	12.8	128.3	0.94
630	66	0.78	0.404	0.30	317	31.7	52.0	1.48
1000	84	1.26	0.654	0.38	514	51.4	32.1	1.89
2000	119	2.54	1.313	0.53	1031	103.1	16.0	2.67

The recommended build is the 1000 V class: **84 turns** of AWG16 on a 70 mm former - L = 1.26 mH, tuned with 32 nF, driven with under 1.9 V, dissipating 4.7 W, and delivering a **51 V tuned RX signal** - comfortable for any comparator, with the 91.7 % wheel dip riding on top of it.

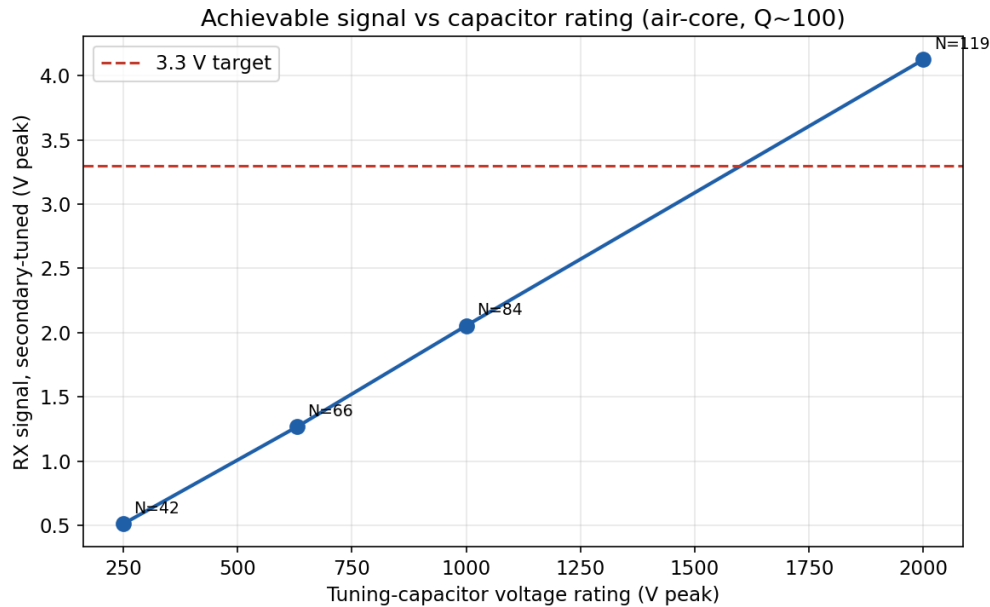


Figure 8 - The air-core design table visualised.

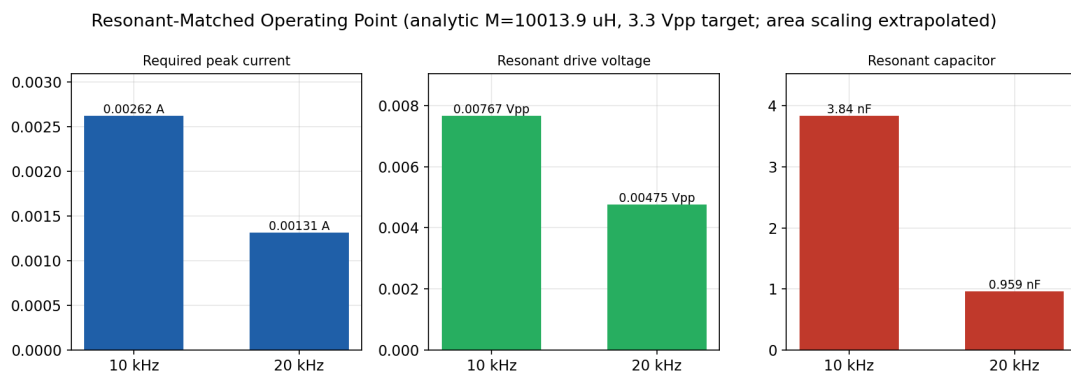


Figure 9 - Resonant-matched operating point for the large-coil analytic design at 10 vs 20 kHz, computed live from the verified anchor.

7.3 The earlier flux-loss studies

The sweep_analysis notebook parameterises the problem differently - assumed flux-loss percentages rather than a FEM anchor - and produced the current/voltage/gauge trade-off figures (05-08 in reports/figures/). They remain useful intuition for how brutally the required drive current grows as coupling efficiency falls toward 99.99 % loss, and their internal resonance arithmetic is checked by the sanity suite; but the FEM-anchored chain above supersedes them for absolute design numbers.

8. How to read every data file

Everything the pipeline produces lands in reports/. Column-by-column:

- **coil_parameter_sweep_femm.csv** - the anchor sweep. Turns, drive current, solved TX current, RX flux linkage (Wb), M (μH), RX voltage. The 100-turn row must equal config.M0_UH; the sanity suite enforces this.
- **frequency_sweep_femm.csv** - per-frequency M, RX voltage and TX self-inductance L_TX (from TX flux / TX current).
- **wheel_dip.json / wheel_dip_result.txt** - baseline and wheel-present M, RX voltages, dip percentage.
- **doe_rsm_result.json** - every DOE run's (shift, tilt) $\rightarrow$ M, the fitted response-surface coefficients, per-axis diagnostics (interior maximum / clamped / fell back), the infeasible-point list, and the verified optimum.
- **scaled_geom_result.json** - the optimum shift applied, the largest feasible scale factor, and M before/after; the geometry itself is femm/optimal_geom_scaled.fem.
- **optimal_design.json** - the section-7 design table plus the recommended build with all constants (gauge, Q, skin depth, capacitor).
- **coil_parameter_sweep.csv** - the analytic turns/area grid (anchored, extrapolated area term).
- **axle_counter_sweep_data.csv** - the notebook's flux-loss parameterisation (frequency, turns, loss %, gauge $\rightarrow$ current, voltages, capacitor).
- **femm_live_result.txt** - the DC-vs-AC validation transcript; look for DONE OK.
- **sanity_check_report.md / RUN_SUMMARY.md** - the audit trail and the one-page run summary; figures/ holds the 13 numbered PNGs, all regenerable from the raw .fem/.ans files.

9. The sanity-check suite: how the numbers police themselves

sanity_check.py is the folder's verification gate - basic electromagnetics only, no FEMM results assumed. It runs in two layers. **Layer 1** builds a reference coil from first principles (30/40 mm radii, 25 mm length, 70 % packing $\rightarrow$ 212 turns of 18 AWG) and derives L by Wheeler's multilayer formula, AC resistance, Q, tuning capacitance and a free-space dipole estimate of coil-to-coil M. It then *solves that same coil in FEMM* (clean axisymmetric model, no rail): **FEMM 3.180 mH vs Wheeler 3.267 mH - 2.7 % apart** - validating units, circuit-turns handling and boundary setup against the textbook.

Layer 2 audits every report file against identities that must hold regardless of what produced them: Faraday's $V = 2\pi f \cdot \Phi$ on every AC row of every CSV ($\leq 1\%$); M independent of drive current ($< 0.005\%$); the N^2 law ($\leq 1.8\%$); resonance identities $C = 1/((2\pi f)^2 L)$ and $V_{\text{cap}} = I/(2\pi f C)$ in both design tables; dip arithmetic; DOE optimum inside the sampled grid; the model depth matching config; and the free-space dipole (8.4 μH at 121 mm separation) upper-bounding the rail-shielded FEMM value (0.93 μH) - the rail can only take flux away. **Current status: 28 passed, 0 failed, 0 skipped.** Run it any time with `python sanity_check.py`; it exits non-zero on any failure, so it can gate CI or the .bat wrappers.

10. Honest caveats and open items

- **The 25 mm depth is an assumption.** It matches the sanity-check reference coil, not a measured part. Absolute fluxes and voltages scale linearly with it; set config.COIL_DEPTH_MM to the real coil's axial length and rerun (one command per solver) when the hardware is fixed.
- **The area-scaling law is unvalidated.** $M \propto \text{area-per-coil}$ at fixed separation is an extrapolation; at 50 $\times$ per-coil scale it strains belief even though the resulting $k \sim 0.15$ is not unphysical. Validate by measurement or a dedicated FEM study before building the big-coil design points.
- **2D planar limits.** The model captures the rail cross-section perfectly but ignores end effects, coil curvature out of plane, and wheel approach along the track. A 3D study (or bench measurement) is the natural next

step for the dip waveform vs wheel position.

- **Q = 100 is assumed** for the tuned RX signal in the design table - realistic for solid-wire air coils at 25 kHz, but proximity effects in a tight winding can lower it; litz wire raises it.
- **The optimum sits against the mechanical clearance limit.** The RSM wanted more shift and tilt than the rail allows, so the reported optimum is a boundary point; a finer grid between -4 and -8 mm with collision-aware spacing would sharpen it.
- **Legacy artefacts.** The flux-loss notebook figures and axle_counter_sweep_data.csv predate the verified anchor; they are kept for intuition and their internal arithmetic is sanity-checked, but the FEM-anchored chain supersedes them.

With those caveats stated: every number in this report traces to a fresh solve or to an identity the sanity suite verifies on demand, the two silent FEMM pitfalls (auto-save on analyze, selection clearing) are engineered out of the codebase, and the pipeline rebuilds everything - data, figures, this report's inputs - headlessly from a clean checkout.